Data Analysis and Visualization

Fitness topic Presentation

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Introduction

Current Landscape: With the growing focus on metabolic health, the rise of wearable technology provides unique opportunities to track our wellness.

Our Vision: By leveraging the power of Data Science, we reveal the complex patterns within exercise physiology, transforming complexity into understanding and creating pathways for transformative health insights.

Research Questions & Data

Research Questions:

- Does Gender significantly affect Workout Type selection?
- Which factors most strongly influence Calories Burned?

Data Overview

- Source: **Kaggle "Life Style Data".**
- The dataset consists of approximately 20,000 observations and 54 variables, sourced from reputable fitness tracking applications, providing essential insights for analyzing workout trends and behaviors.

Overview

Methodology

EXPLORATORY DATA ANALYSIS (EDA)

Research Quesion 1

Research Quesion 2

Distribution Testing and Hypothesis Testing

Correlation Analysis and Multiple Linear Regression

Conclusion

EXPLORATORY DATA ANALYSIS (EDA)

Data Overview

First, we begin with Exploratory Data Analysis to enhance our understanding of the dataset.

- We selected just 15 variables rather than attempting to utilize all available ones, adhering to the principle of 'quality over quantity' in lifestyle research.
- Among the 20,000 records, no missing values were detected in the chosen data columns.
- Additionally, there were no duplicate records within the 20,000 observations.

	Column	Data Type
0	Age	float64
1	Gender	object
2	Weight (kg)	float64
3	Height (m)	float64
4	BMI	float64
5	Fat_Percentage	float64
6	Max_BPM	float64
7	Avg_BPM	float64
8	Resting_BPM	float64
9	Session_Duration (hours)	float64
10	Calories_Burned	float64
11	Workout_Type	object
12	Workout_Frequency (days/week)	float64
13	Water_Intake (liters)	float64
14	Experience_Level	float64

```
Numerical variables (13):
  1. Age
  2. Weight (kg)
  3. Height (m)
  4. BMI
  5. Fat_Percentage
  6. Max_BPM
  7. Avg_BPM
  8. Resting_BPM
  9. Session_Duration (hours)
  10. Calories_Burned
  11. Workout_Frequency (days/week)
  12. Water_Intake (liters)
  13. Experience_Level

Categorical variables (2):
  1. Gender
  2. Workout_Type
```

EXPLORATORY DATA ANALYSIS (EDA)

Descriptive Statistics

Descriptive Statistics for All Numerical Variables

	count	mean	std	min	25%	50%	75%	max	variance	skewness	kurtosis
Age	20000.00	38.85	12.11	18.00	28.17	39.86	49.63	59.67	146.76	-0.09	-1.20
Weight (kg)	20000.00	73.90	21.17	39.18	58.16	70.00	86.10	130.77	448.30	0.77	-0.03
Height (m)	20000.00	1.72	0.13	1.49	1.62	1.71	1.80	2.01	0.02	0.33	-0.70
BMI	20000.00	24.92	6.70	12.04	20.10	24.12	28.56	50.23	44.91	0.79	0.81
Fat_Percentage	20000.00	26.10	5.00	11.33	22.39	25.82	29.68	35.00	24.96	0.08	-0.71
Max_BPM	20000.00	179.89	11.51	159.31	170.06	180.14	189.42	199.64	132.50	-0.03	-1.18
Avg_BPM	20000.00	143.70	14.27	119.07	131.22	142.99	156.06	169.84	203.57	0.09	-1.18
Resting_BPM	20000.00	62.20	7.29	49.49	55.96	62.20	68.09	74.50	53.13	-0.06	-1.17
Session_Duration (hours)	20000.00	1.26	0.34	0.49	1.05	1.27	1.46	2.02	0.12	0.02	-0.33
Calories_Burned	20000.00	1280.11	502.23	323.11	910.80	1231.45	1553.11	2890.82	252233.95	0.68	0.28
Workout_Frequency (days/week)	20000.00	3.32	0.91	1.94	2.98	3.01	4.00	5.06	0.83	0.15	-0.80
Water_Intake (liters)	20000.00	2.63	0.60	1.46	2.17	2.61	3.12	3.73	0.37	0.06	-1.04
Experience_Level	20000.00	1.81	0.74	1.00	1.01	1.99	2.02	3.05	0.54	0.33	-1.11

- **Robust Data Foundation:**
  - Total Observations: 20,000 (100% Complete).
  - Target Population: Mature Adults (Avg. Age 39).
- **High-Intensity Activity Profile:**
  - Avg. Session: 1.26 hours.
  - Avg. Calories Burned: 1,280 kcal (High variance indicates diverse workout types).
  - Fitness Level: Good (Resting HR ~62 bpm).
- **Distribution Insights (For Modeling):**
  - Physiological metrics (Heart Rate, Age) follow a Normal Distribution.
  - Body Composition (Weight, BMI) is Right-Skewed,

EXPLORATORY DATA ANALYSIS (EDA)

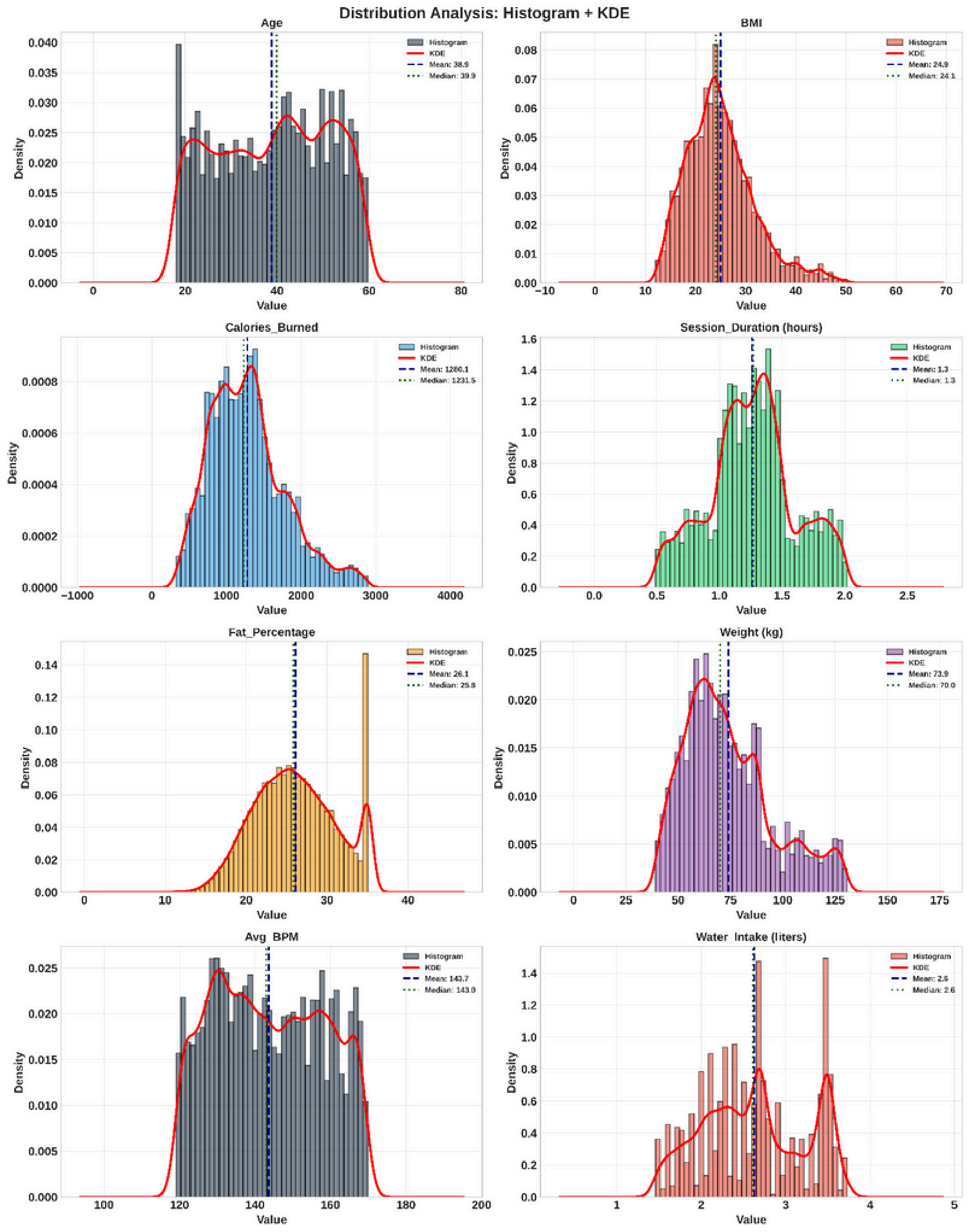
Descriptive Statistics

Statistical Metrics & Data Interpretation

1. **Central Tendency (The Center)**
   - Observation: Mean > Median.
   - Insight: Data has a Positive Skew. The average is inflated by high-performing outliers, making Median the more robust metric.
2. **Variability (The Spread)**
   - Observation: High Standard Deviation.
   - Insight: High Heterogeneity. The dataset represents a diverse population (from lean to heavy), not a uniform group.
3. **Distribution Shape (The Balance)**
   - Skewness (+): Right-skewed distribution driven by a minority of "super-users".
   - Kurtosis (High): Heavy tails indicate the presence of significant extreme outliers.

EXPLORATORY DATA ANALYSIS (EDA)

Comprehensive Data Visualization (Histogram + KDE)



1. Physiological Skewness (Right-Skewed Distribution)

- Variables: BMI, Weight, Calories Burned.
- Observation: These metrics exhibit a positive skew with a long tail to the right.
- Implication: This reflects biological reality where a subset of the population (outliers) has significantly higher body mass or exercise intensity than the average.

2. Behavioral Rounding (Multimodal Distribution)

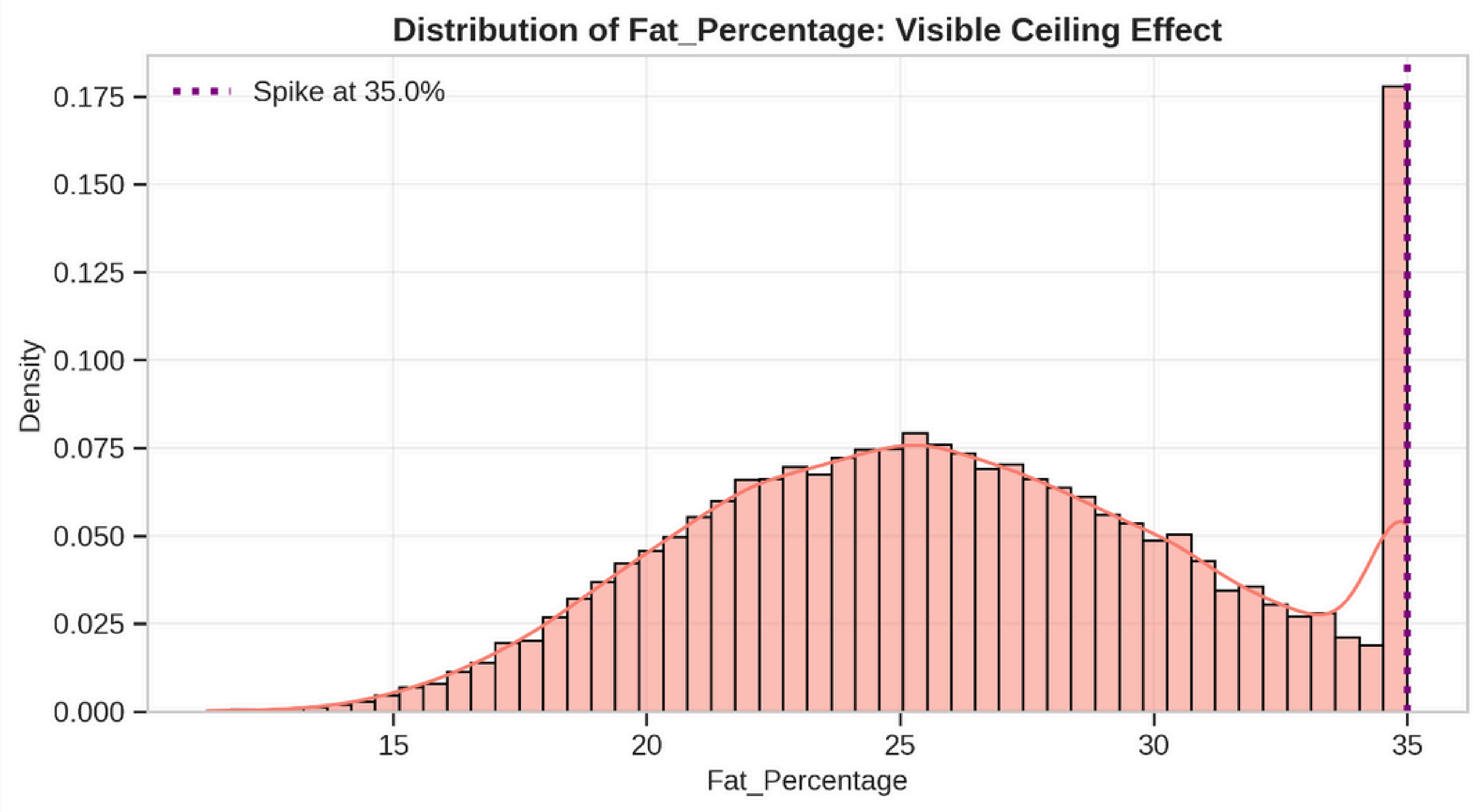
- Variables: Water_Intake, Session_Duration.
- Observation: Distinct multimodal peaks appear at integer or half-integer intervals (e.g., 2.0L, 3.0L water; 1.0h, 1.5h duration).

3. Data Anomaly (Ceiling Effect)

- Variables: Fat_Percentage.
- Observation: An unnatural spike occurs strictly at the 35% mark.

EXPLORATORY DATA ANALYSIS (EDA)

Handle Data Anomaly



The histogram reveals an unnatural accumulation of data points exactly at . This pattern strongly suggests a “Ceiling Effect” (Right-Censoring), implying the measurement instrument was limited to , or data collection protocols grouped all higher values into this cap.

EXPLORATORY DATA ANALYSIS (EDA)

Handle Data Anomaly

To determine if this is a data error or a valid representation of high-risk individuals, we perform a statistical comparison:

- Group A (Capped): Individuals with .
- Group B (Normal): The rest of the population.
- Metric: We compare their Mean BMI and Weight to see if Group A significantly differs from the general population.

	Metric	Capped Group (Fat=35%)	Normal Group (Rest)	Difference
0	Mean BMI	39.69	23.72	15.97
1	Mean Weight (kg)	114.26	70.61	43.65
2	Mean Age	39.40	38.81	0.60
3	Workout Freq (days/week)	3.05	3.34	-0.29

Conclusion:

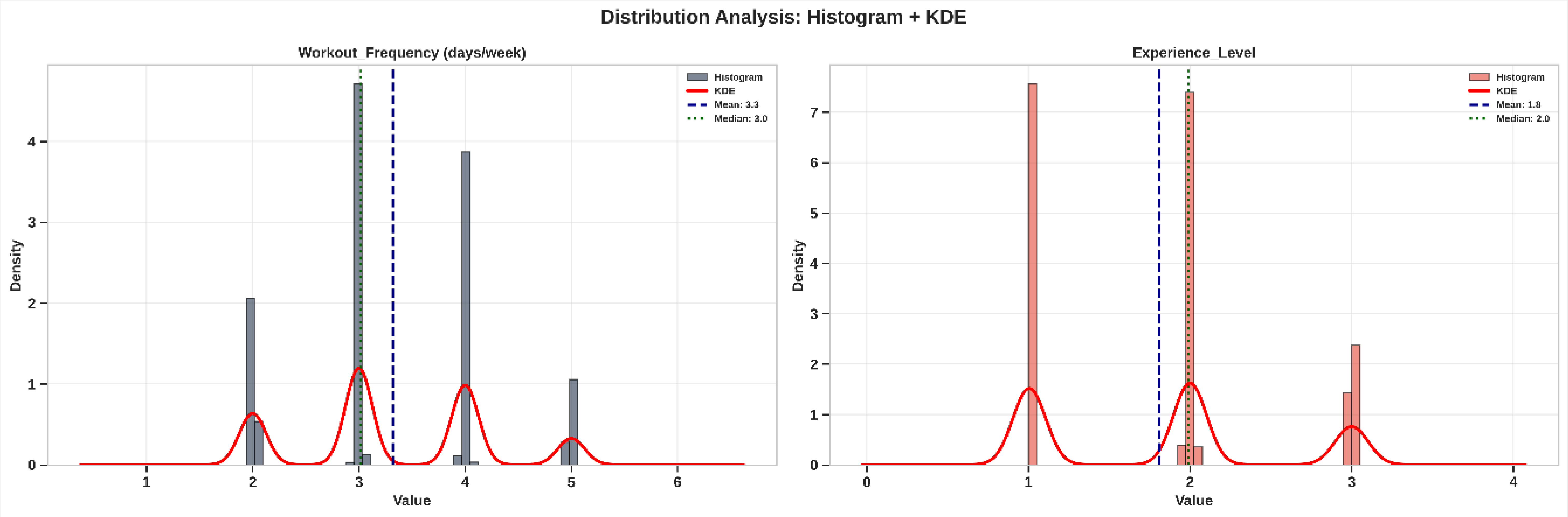
- The ‘Capped Group’ has a Mean BMI of 39.69, which is classified as severe obesity.
- Compare this to the ‘Normal Group’ BMI of 23.72 (Normal/Overweight range).
- The Capped Group is on average 43.65 kg heavier than the rest.

=> **Final decision: Retain data.**

Removing these rows would bias the study by excluding the highest-risk population.

EXPLORATORY DATA ANALYSIS (EDA)

Comprehensive Data Visualization (Histogram + KDE)



Observation: While analyzing numerical distributions, we detected anomalous patterns in Experience_Level and Workout_Frequency.

The Issue: Although stored as continuous floats (e.g., 1.99, 3.01), the histograms exhibit distinct, separated peaks rather than a smooth curve.

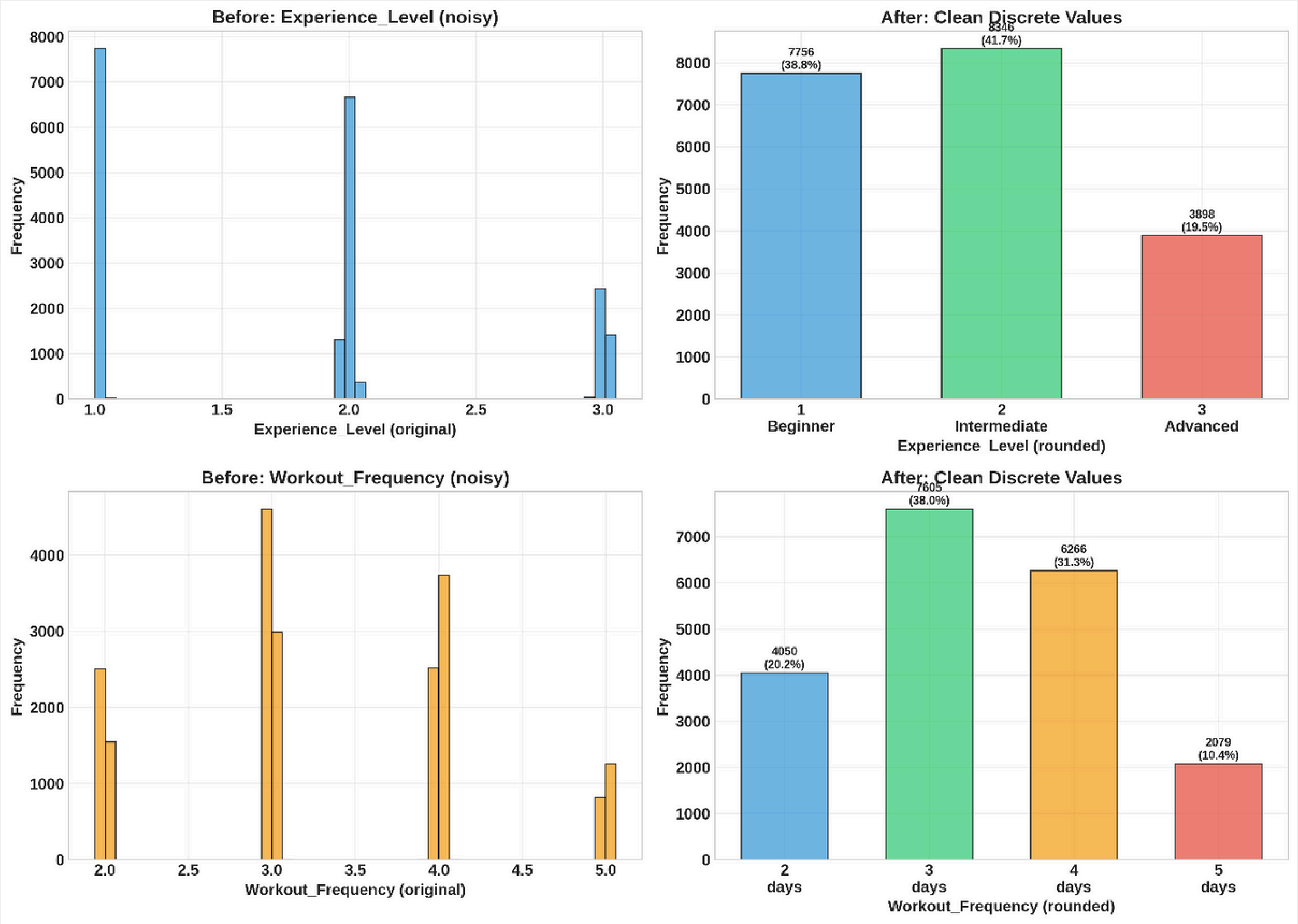
Diagnosis: These are inherently Discrete/Ordinal variables affected by measurement noise or floating-point artifacts.

EXPLORATORY DATA ANALYSIS (EDA)

Handle Data Anomaly

Action Steps

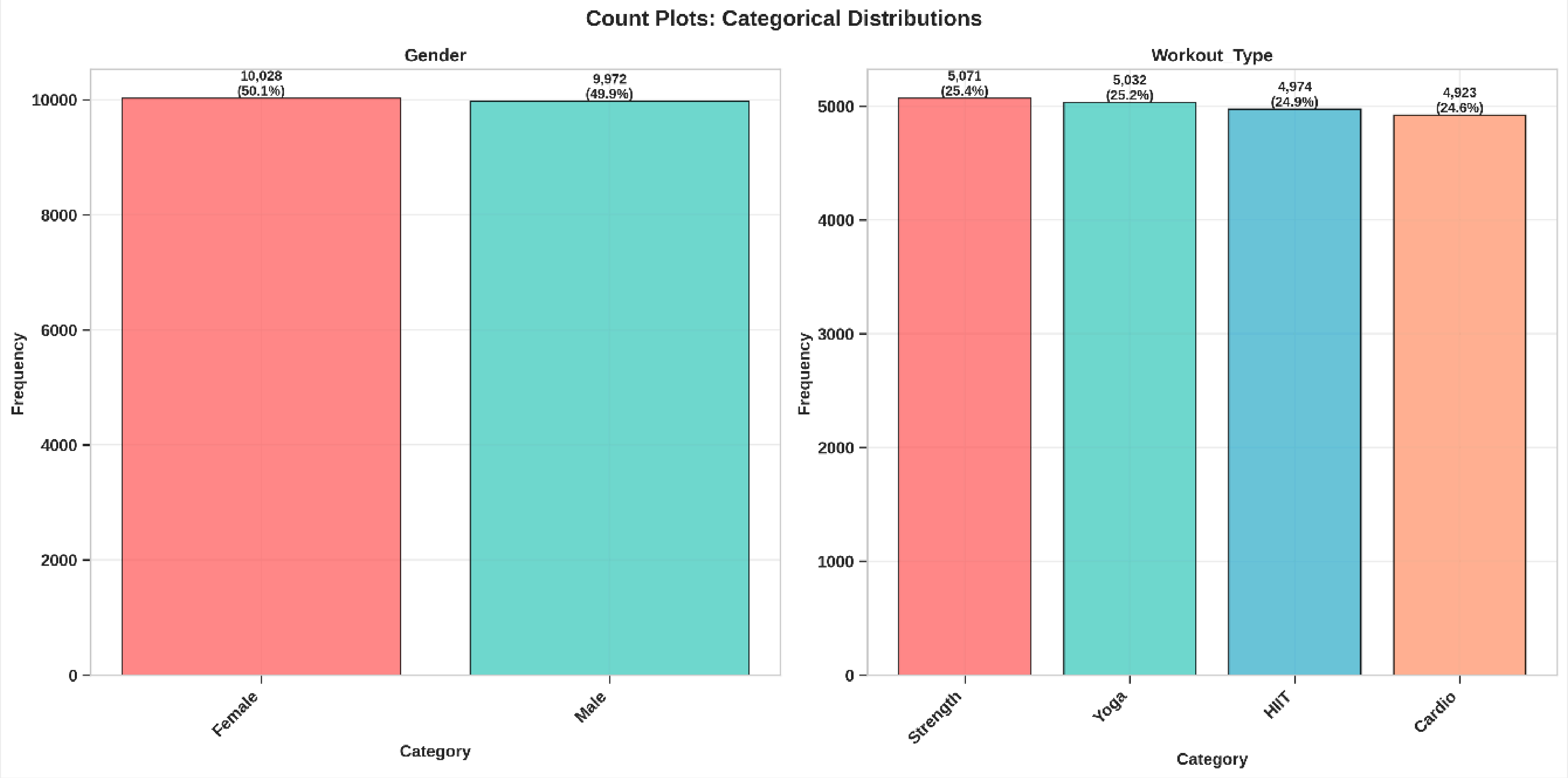
- **Clean Data: Round values to the nearest integer.**
- Workout Frequency: Grouped into four clear categories:
  - 2 (days/week)
  - 3 (days/week)
  - 4 (days/week)
  - 5 (days/week)
- Experience Level: Organized into three distinct levels:
  - 1 (Beginner)
  - 2 (Intermediate)
  - 3 (Advanced)



EXPLORATORY DATA ANALYSIS (EDA)

Comprehensive Data Visualization

Gender Distribution **Workout Type Distribution**

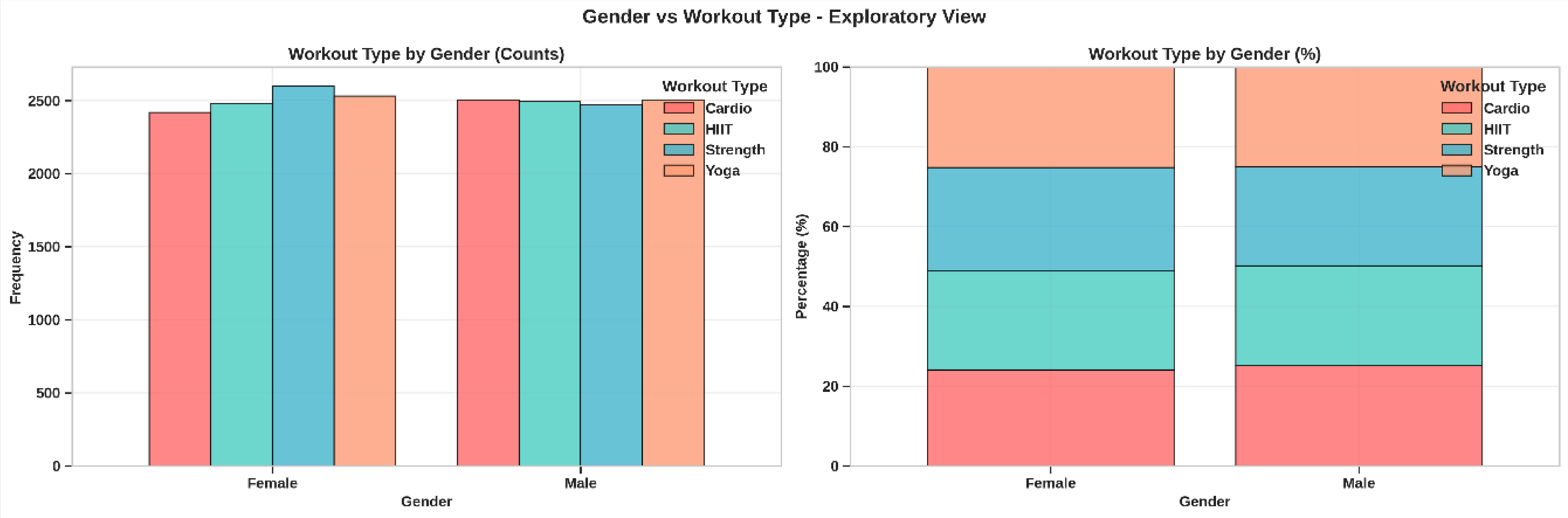


Key Observations:

- **Gender Parity:**
  - Result: A near-perfect 50/50 split (Females: 50.1% vs. Males: 49.9%).
  - Insight: No gender bias exists. This ensures that any difference we find in workout preference is real, not due to having more men or women in the sample.
- **Uniform Workout Distribution**:
  - Result: All 4 workout types (Strength, Yoga, HIIT, Cardio) are equally represented (~25% each).
  - Insight: We have a Uniform Distribution. The dataset does not favor any specific exercise, providing a fair ground for comparison.
- **Conclusion for Analysis:**
  - Zero Class Imbalance: The absence of imbalance means we don't need resampling techniques (like SMOTE) for classification models later on. Ideally suited for statistical testing (Chi-Square).

EXPLORATORY DATA ANALYSIS (EDA)

Examines the relationship between Gender and Workout Type



Observation: The chart reveals the distribution of workout preferences across genders. If the bars show similar heights for males and females across workout types, it demonstrates that there is no significant gender bias in the dataset (data is balanced regarding gender preferences)

EXPLORATORY DATA ANALYSIS (EDA)

Outlier Detection and Handling

1. The Detection Process

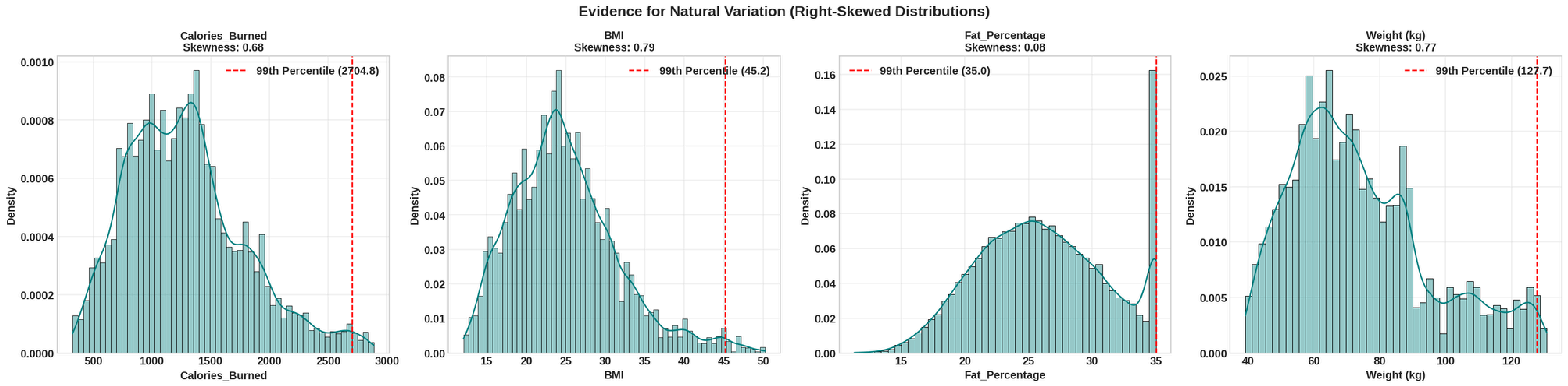
- Methodology
  - Utilized a “Multi-Method Consensus” strategy to reduce the occurrence of false positives.
- Input: Incorporated three algorithms:
  - IQR
  - Z-Score
  - Isolation Forest
- Rule
  - An item is flagged only if the vote is at least 2/3 from the methods used.

```
==================================================================
          OUTLIER DETECTION SUMMARY (VOTING STRATEGY >= 2 METHODS)
==================================================================
Weight (kg)                         :   83 outliers detected (0.41%)
BMI                                 :  227 outliers detected (1.14%)
Fat_Percentage                      :    1 outliers detected (0.01%)
Calories_Burned                     :  145 outliers detected (0.73%)
```

EXPLORATORY DATA ANALYSIS (EDA)

Outlier Detection and Handling

- **2. The Decision**
- **Observation**
  - Histograms reveal continuous distributions that are smooth and right-skewed, lacking any unnatural gaps.
- **Biological Plausibility**
  - Elevated values signify legitimate subgroups rather than errors:
    - High BMI (>35): Indicates Obesity Class II/III.
    - High Caloric Intake (>2500): Corresponds to high-performance athletes.
- **Final Decision**
  - ***<u>RETAIN ALL DATA.</u>***
- **Justification**
  - Eliminating this data would lead to bias and compromise vital health insights related to obesity and athletic performance.



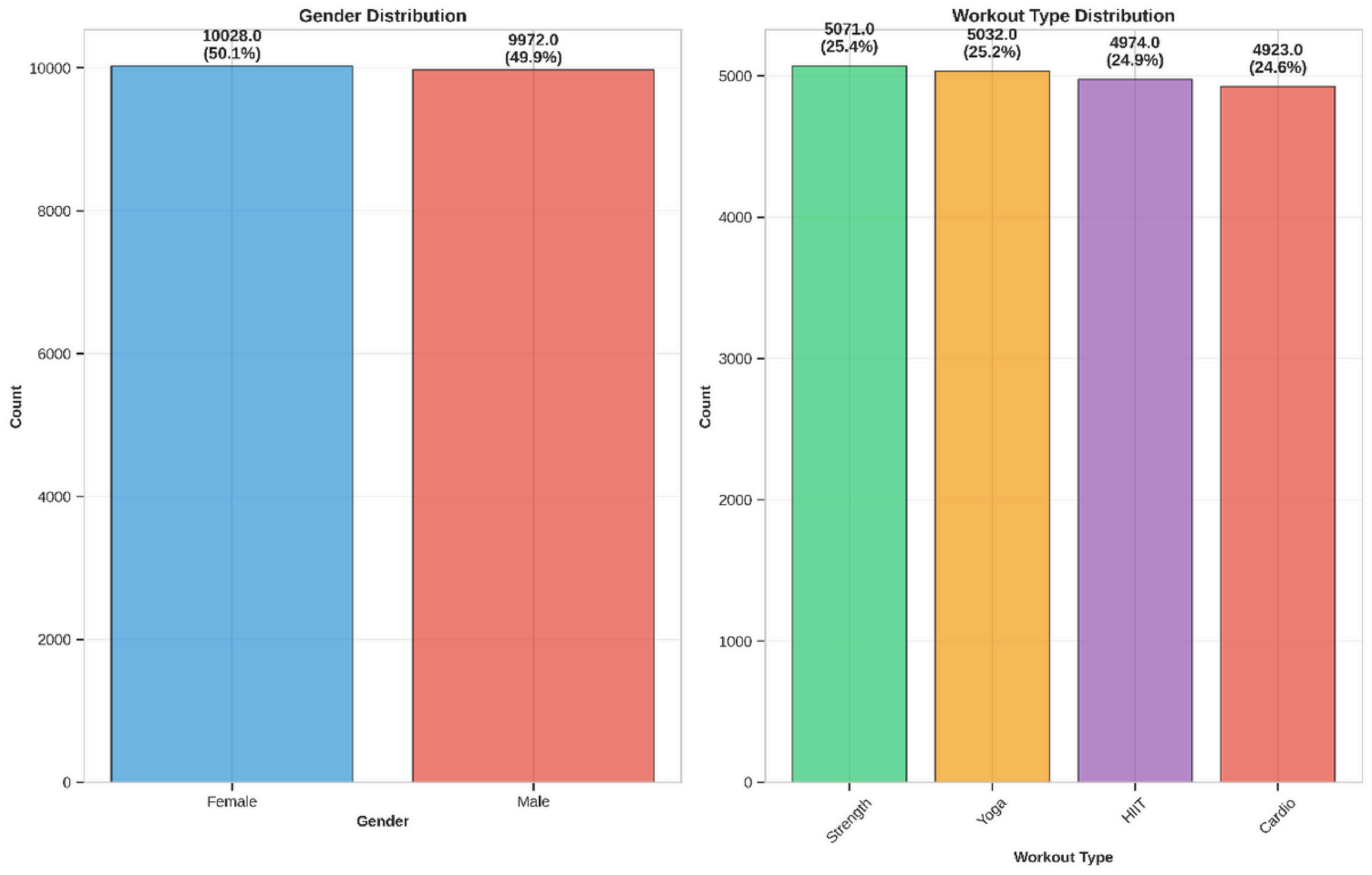
Research Question 1

Does Gender significantly affect Workout Type selection?

- **Method: (Distribution Testing, Hypothesis Testing)**
- Variables: Gender (Male/Female), Workout Type (Cardio/HIIT/Strength/Yoga)
- Hypotheses:
  - H_0: Gender and Workout Type are independent
  - H_1: Gender and Workout Type are dependent
- Context: Validating common stereotypes (e.g., "Males prefer Strength, Females prefer Yoga").
- Analytical Roadmap
  - Step 1: Distribution Testing
  - Step 2: Hypothesis Testing

Distribution Testing

Visual Inspection

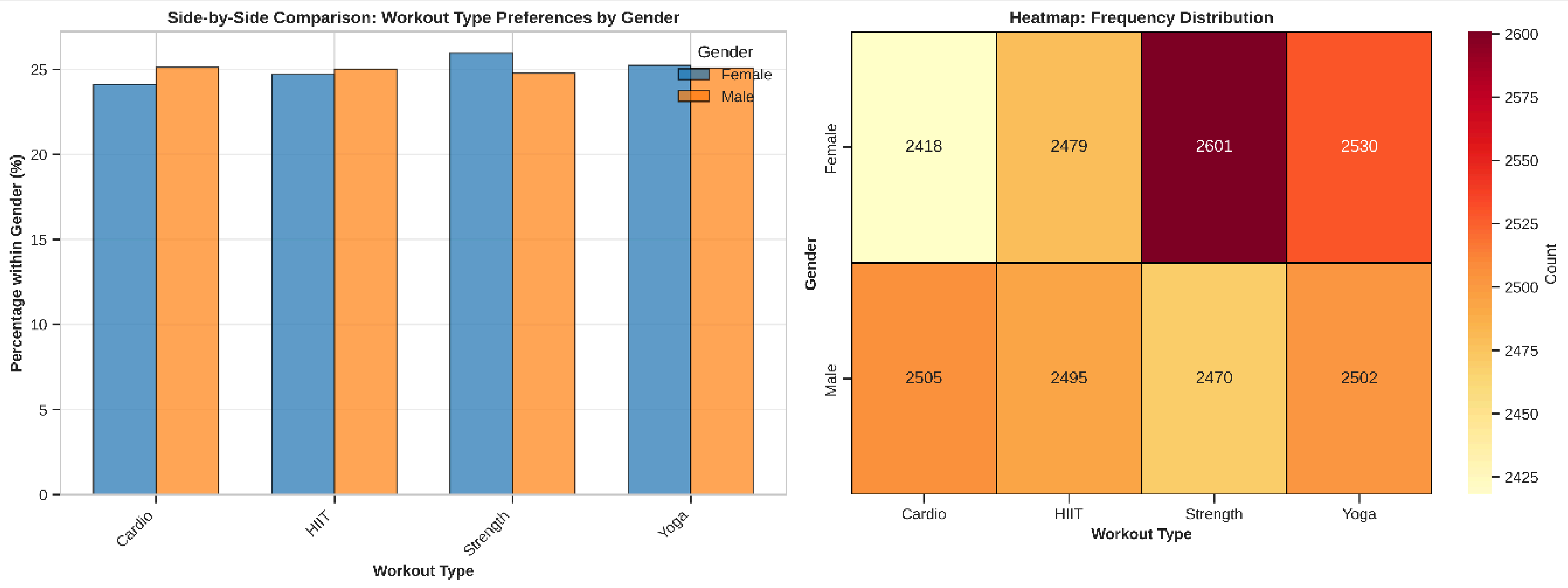


Observation

- Gender: Nearly perfect symmetry
- Workout Type: Even distribution across four categories
- The heights of the bars remain consistent, showing no noticeable skew or sampling bias.

Distribution Testing

Visualizing Relationships (Bivariate Analysis)



- **Side-by-Side Comparison (Left Chart):**
  - Pattern: The blue (Female) and orange (Male) bars are nearly equal in height across all workout types.
  - Insight: Visual evidence suggests Independence. No gender dominates any specific exercise category.
- **Heatmap Distribution (Right Chart):**
  - Intensity: Colors are relatively uniform (mostly orange/yellow tones).
  - Detail: Counts range narrowly (2418 - 2601), confirming a balanced distribution without extreme clusters.

Distribution Testing

Uniformity Test

```
--- Uniformity Test: Gender ---
Observed Counts:
Gender
Female    10028
Male       9972
Name: count, dtype: int64
Chi-square Statistic: 0.1568
P-value: 0.6921
--- Uniformity Test: Workout_Type ---
Observed Counts:
Workout_Type
Strength    5071
Yoga        5032
HIIT        4974
Cardio      4923
Name: count, dtype: int64
Chi-square Statistic: 2.5340
P-value: 0.4692
```

We proceed with a Chi-square Goodness of Fit test to statistically confirm if these variables follow a Uniform Distribution.

- **Method**: Chi-square Goodness of Fit Test (Test for Uniformity)
- **Hypothesis (H0):**The distribution is uniform (equal frequencies).
- **Results**:
    - Gender: Perfectly uniform.
    - Workout Type: Perfectly uniform.

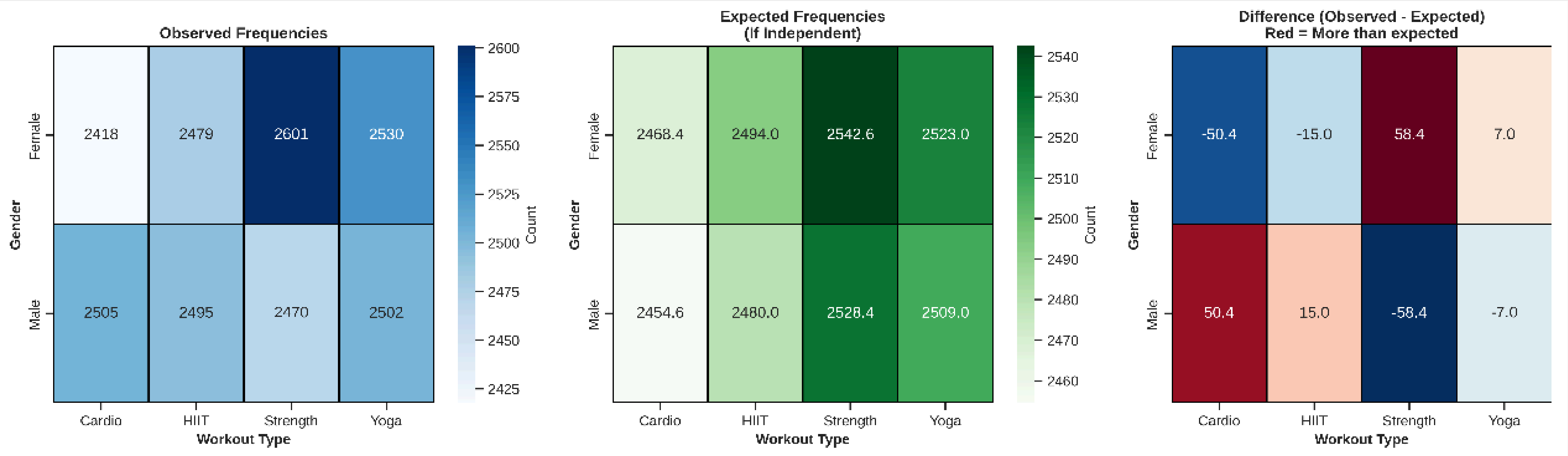
Distribution Testing

Statistical Validation: Chi-Square Assumption Check

This tells us if chi-square test is appropriate for question 1

The Statistical Rule (Cochran's Rule):

- To ensure the Chi-Square test is valid, the Expected Frequency in each cell must be ≥5.
- Why? Counts lower than 5 can cause the test statistic to become unstable and inaccurate.



Conclusion:

- The assumption is strongly satisfied.

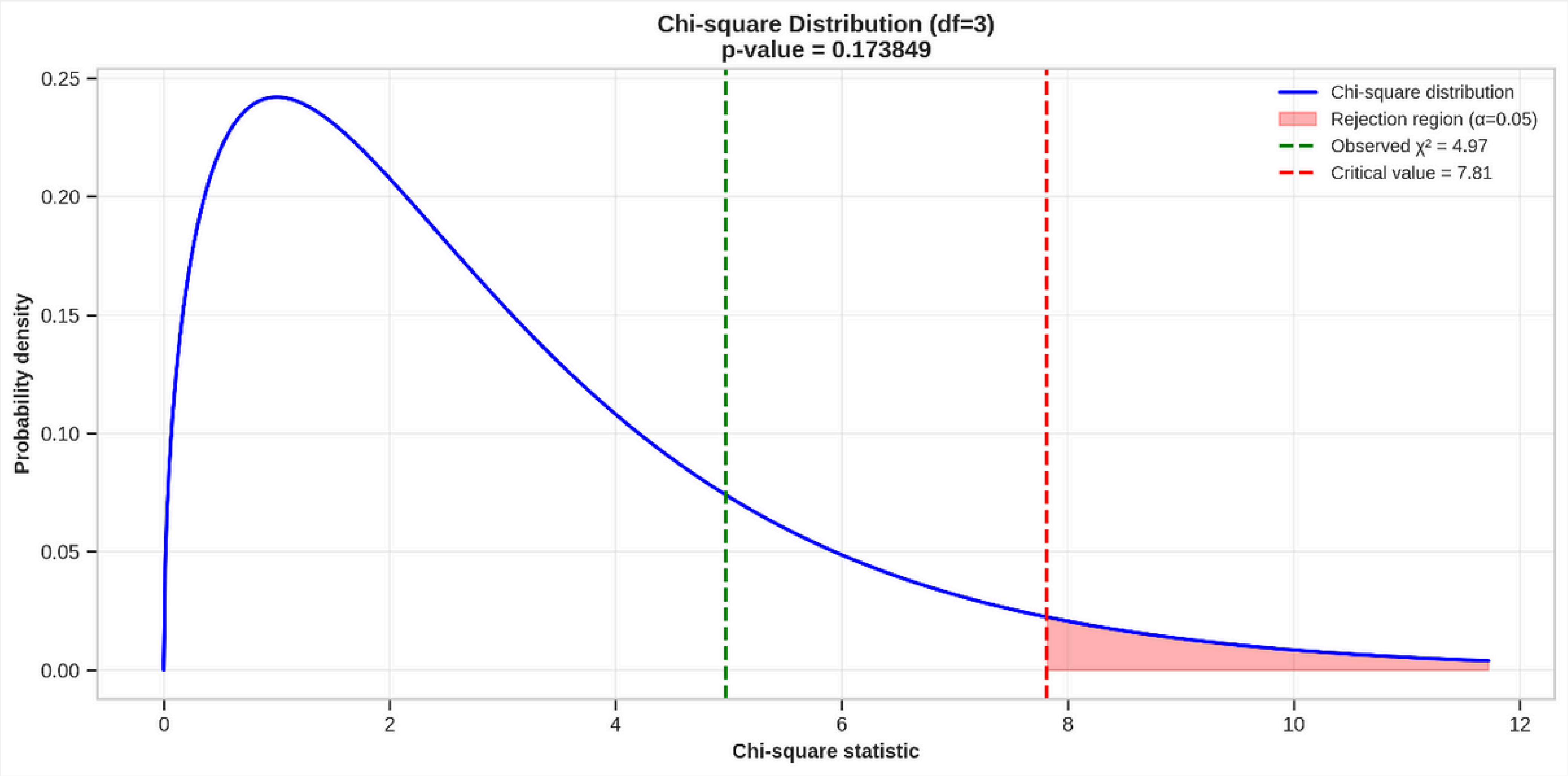
Distribution Testing

Conclusion

- **Chi-square Test assumption:**
    - Independence: Each observation is independent.
    - Expected frequencies: All cells have expected count .
    - Sample size: Sufficient for chi-square test.
- **Preliminary Evidence:**
    - Distribution patterns suggest possible association between Gender and Workout Type.
    - Observed frequencies differ from expected (independent) frequencies.
    - Statistical test needed to determine if differences are significant.
- **Distribution**:
    - Gender and Workout Type follow uniform distributions.

Hypothesis Testing

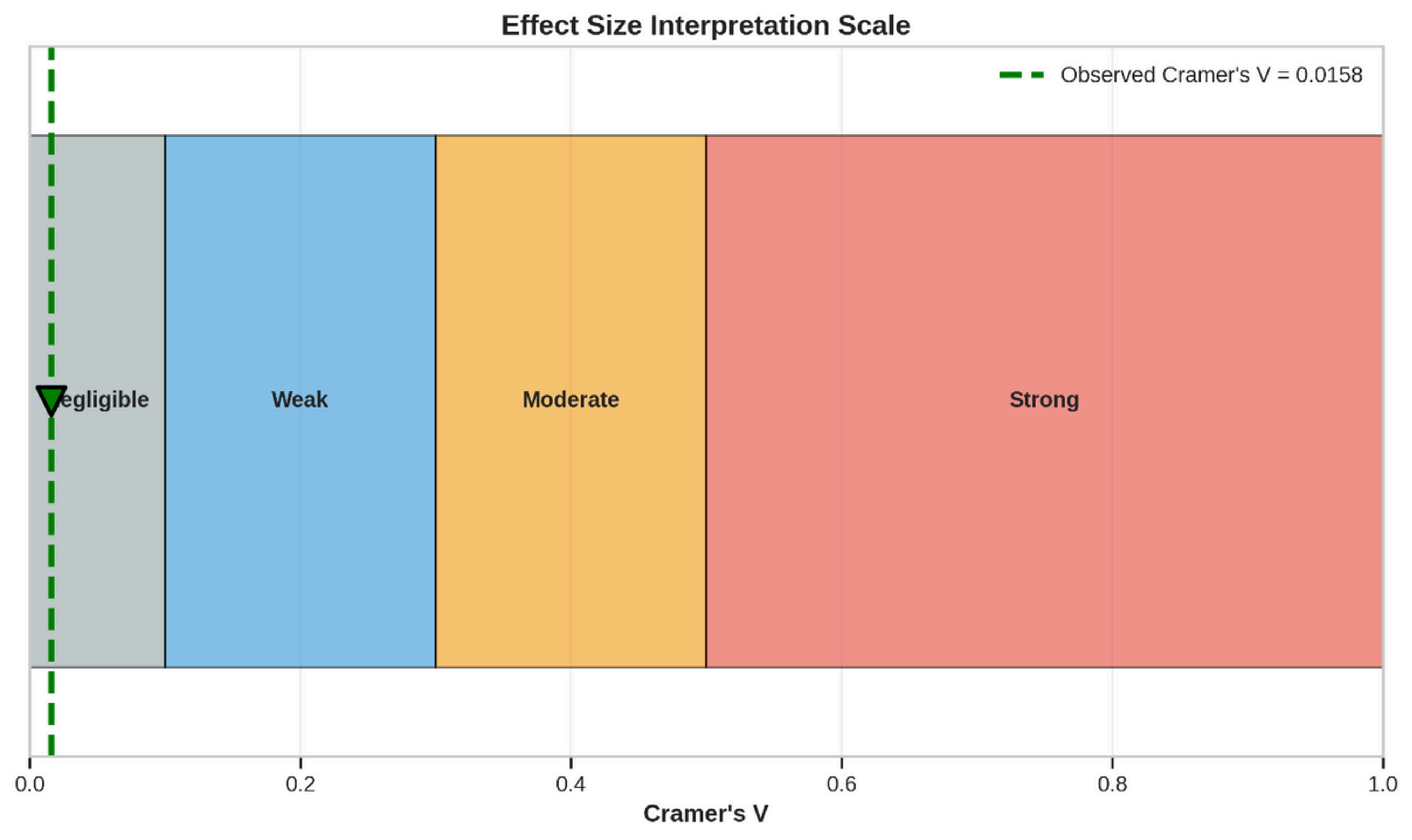
1. **Hypothesis Recap**
    - H0: Independent (No relationship between Gender and Workout Type).
    - H1: Dependent (Relationship exists between Gender and Workout Type).
    - Significance Level=0.05.
2. **Test Statistics**
    - Chi-Square Statistic ($\chi 2$): 4.97
    - Degrees of Freedom (df): 3.
    - P-value: 0.1738.
3. **The Decision**
    - Comparison: P-value(0.1738)>α(0.05).
    - **Conclusion**: **Fail to Reject H0**. There is NO statistically significant association between Gender and Workout Type.



Hypothesis Testing

Deep Dive Analysis (Effect Size & Residual Analysis)

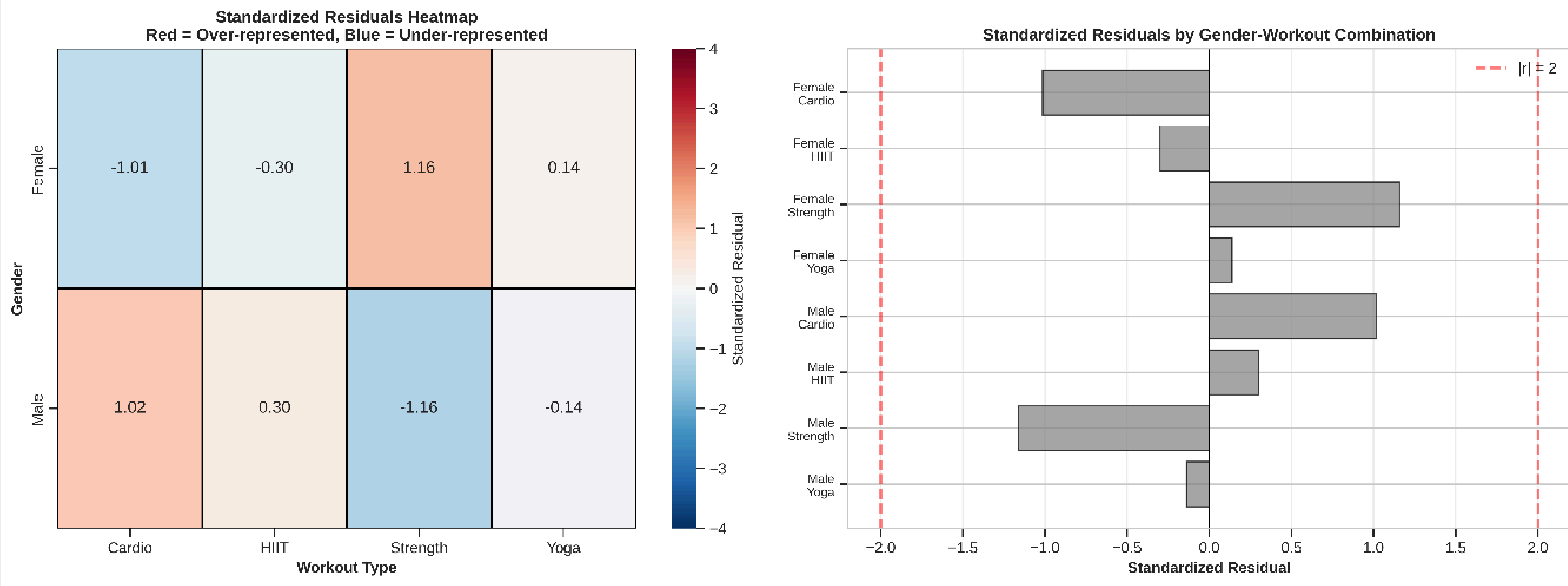
1. **Magnitude of Effect (Cramer's V)**
   - Metric: Cramer's V = 0.0158.
   - Interpretation: "**Negligible**" (< 0.1). The relationship is practically non-existent.
   - Insight: Even if the sample size were larger (forcing significance), the actual impact of Gender on Workout choice is near zero.



Hypothesis Testing

Deep Dive Analysis (Effect Size & Residual Analysis)

Identifying Local Deviations (Standardized Residuals)



- **Analysis:**
  - Range: Residuals fall between -1.16 and +1.16.
  - Threshold: No cell exceeds the critical threshold of ±2.
- Meaning: No specific group (e.g., "Men doing Cardio" or "Women doing Strength") occurs significantly more or less than expected by random chance.

Answer for Research Question 1

Does Gender significantly affect Workout Type selection?

1. **Scientific Conclusion**
    - Result: The hypothesis that "Gender dictates exercise choice" is **REFUTED**.
2. **Practical Implication**
    - Business/Health Insight: Workout programs can be designed gender-neutral. Marketing or recommendation systems should focus on fitness goals (e.g., weight loss vs. muscle gain) rather than gender stereotypes.

Research Question 2

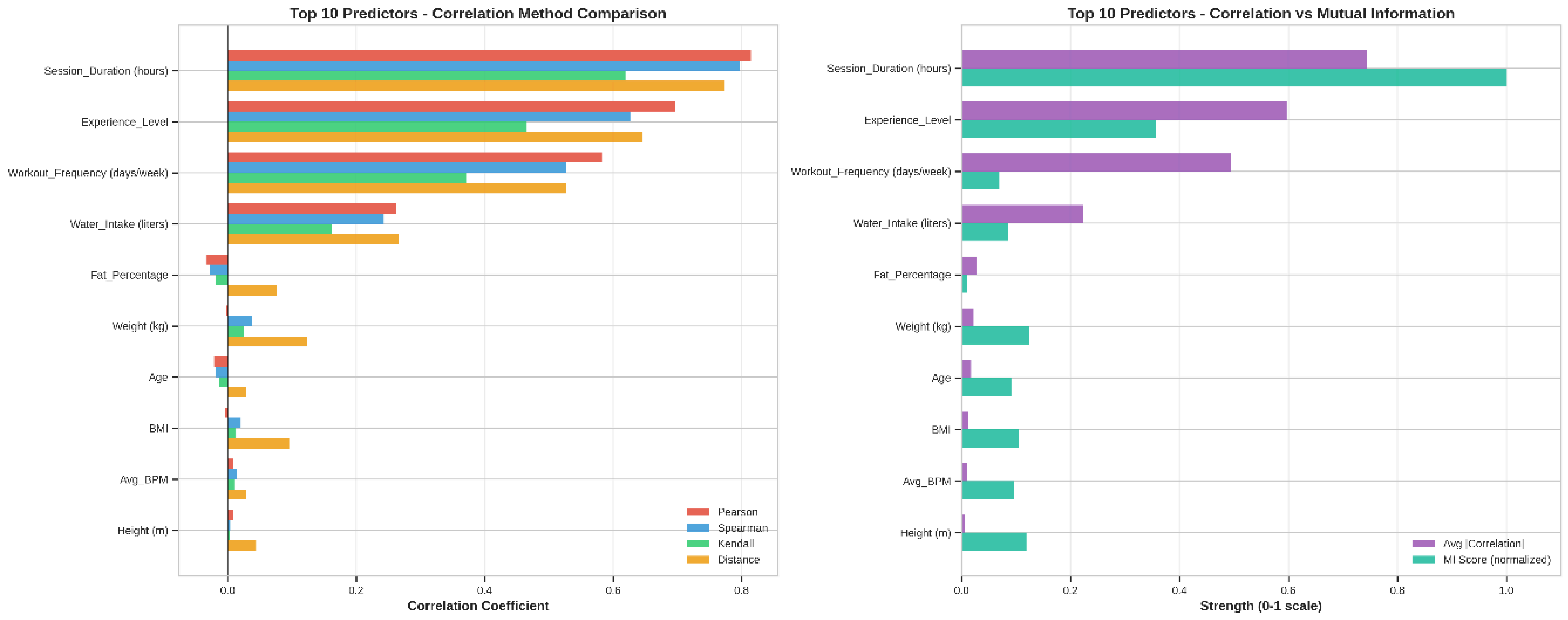
1. **The Research Question**
    - Core Question: Which factors are the strongest predictors of "Calories Burned", and can we accurately estimate energy expenditure?
    - Goal: Build a Predictive Model (Regression) to estimate energy expenditure.
2. **The Analytical Strategy (3-Step Process)**
    - Step 1: Screening (Correlation)
        - Action: Apply 5 Correlation Methods (Pearson, Spearman, Kendall, Distance, Mutual Information).
        - Why? To rank feature importance and capture both linear & non-linear relationships.
    - Step 2: Selection (Feature Engineering)
        - Action: Detect & Remove Multicollinearity (VIF check) and assess Categorical impact (ANOVA).
        - Why? To ensure model stability and remove redundant variables (e.g., Weight vs. BMI).
    - Step 3: Modeling (Regression)
        - Action: Construct a Multiple Linear Regression (MLR) model and validate assumptions.

Correlation Analysis

Identifying Key Drivers via Multi-Method Correlation

Methodology

- Approach: Compared 5 distinct correlation methods (Pearson, Spearman, Kendall, Distance, Mutual Information) to ensure robustness against non-linear relationships.
- Top 10 Predictors (Consensus Results) show in the picture below



Correlation Analysis

Feature Selection & Multicollinearity Handling (Numerical Variables)

1. **The Problem: Information Redundancy**
   - Diagnosis: Severe multicollinearity detected between Weight, Height, and BMI (Pearson r > 0.9).
   - Impact: Including all creates unstable regression coefficients (noise).
2. **The Solution: Strategic Removal**
   - Action 1: Removed Weight and Height, kept BMI (or vice versa) to reduce redundancy while keeping biological info.
   - Action 2: Removed Workout_Frequency due to high correlation with Experience_Level (r = 0.84).
3. **Final Selected Features (Numerical variables)**
   - Numerical: Session_Duration, Experience_Level, Water_Intake.

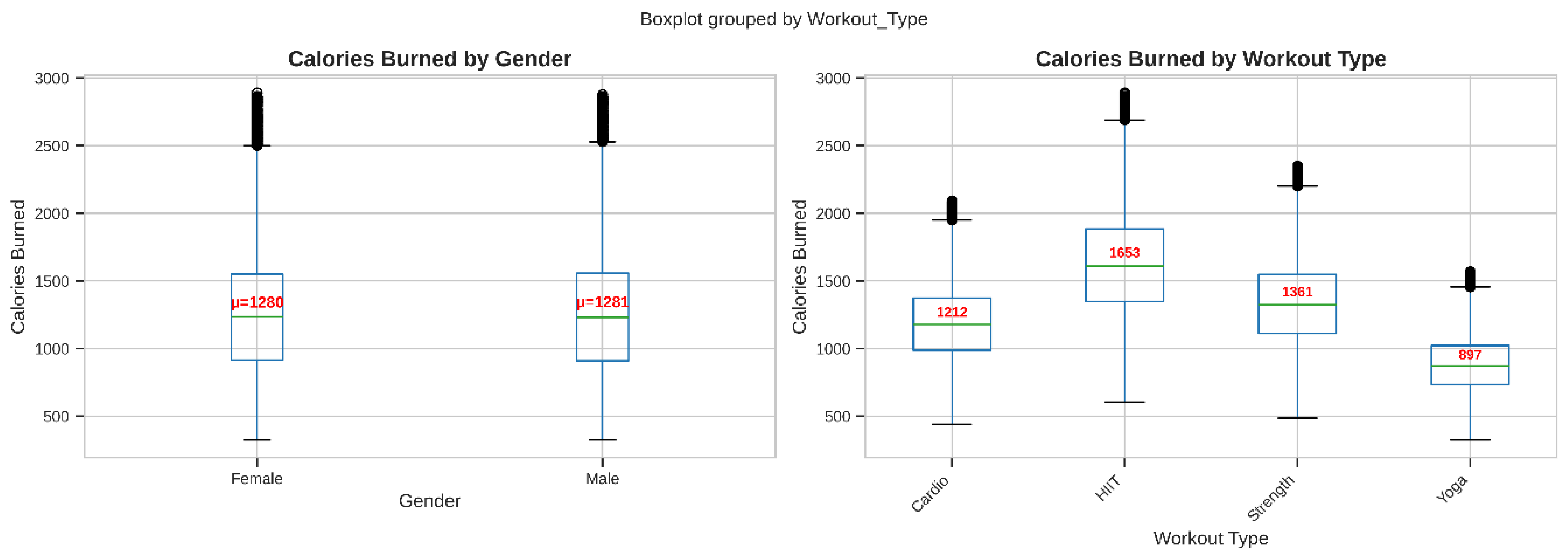
	Variable	Pearson	Spearman	Kendall	Distance_Corr	Avg_Abs_Corr
8	Session_Duration (hours)	0.814	0.797	0.619	0.773	0.744
11	Experience_Level	0.697	0.663	0.543	0.644	0.634
10	Water_Intake (liters)	0.263	0.243	0.163	0.266	0.223

Correlation Analysis

Feature Selection for Categories Variables

Methods

- **Gender Impact**
  - Test: Point-biserial correlation & ANOVA.
  - Decision: exclude Gender .
- **Workout Type Impact**
  - Test: One-way ANOVA.
  - Decision: include Workout_Type.



Observation

- **Gender Impact (Left):**
  - Insight: Gender is insignificant
- **Workout Type Impact (Right):**
  - Insight: Workout type is a strong predictor .

Correlation Analysis

Final Selected Features

- **Numerical:** *Session_Duration, Experience_Level, Water_Intake.*
- **Categorical**: *Workout Type*

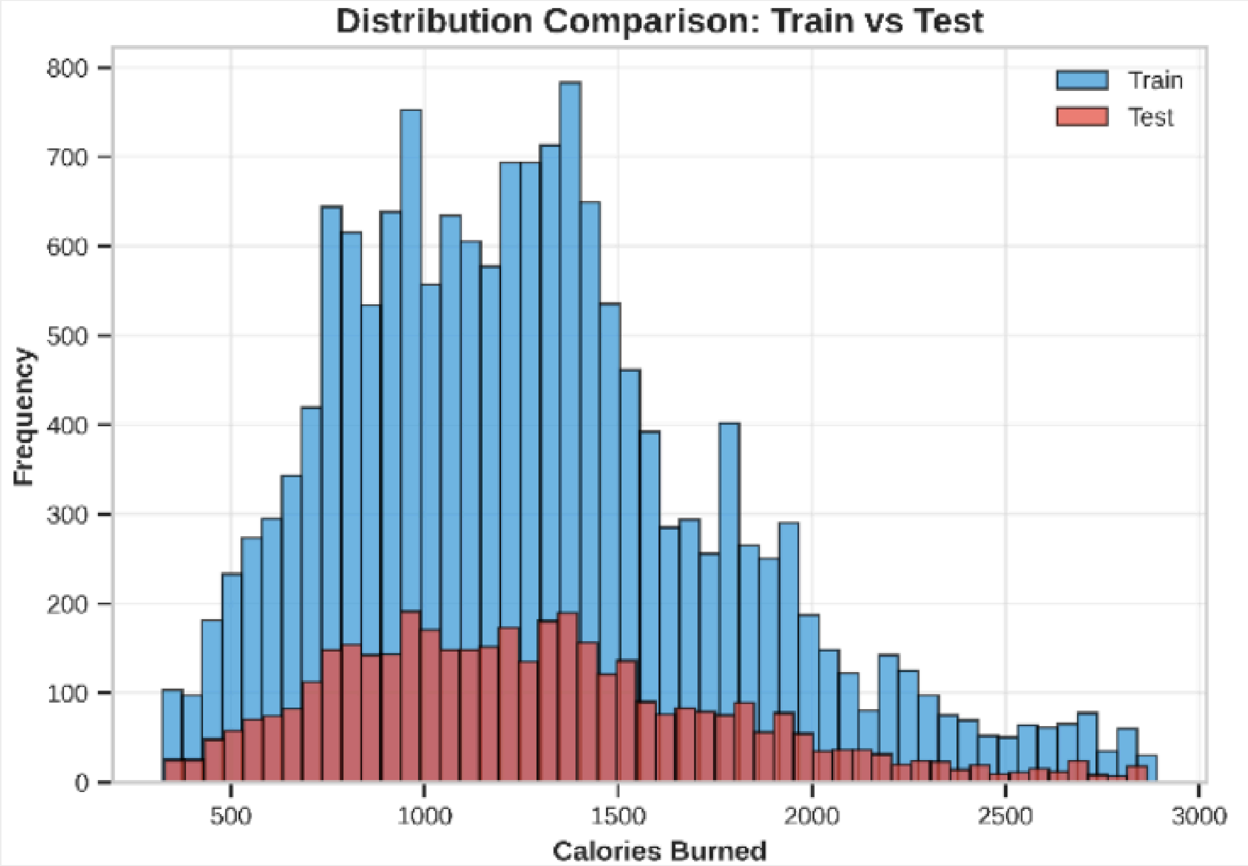
```
A. TOP NUMERICAL PREDICTORS (by consensus across methods):
----------------------------------------------------------------------------------------------------
 #9: Session_Duration (hours)
     Pearson:  0.814  |  Spearman:  0.797  |  Distance:  0.773

 #12: Experience_Level
     Pearson:  0.697  |  Spearman:  0.663  |  Distance:  0.644

 #11: Water_Intake (liters)
     Pearson:  0.263  |  Spearman:  0.243  |  Distance:  0.266


B. CATEGORICAL PREDICTORS ANALYSIS:
----------------------------------------------------------------------------------------------------
 Gender            : ✗ EXCLUDED   | p=0.8822, η²=0.0000 (negligible)
 Workout_Type      : ✓ SELECTED   | p=0.0000, η²=0.2943 (large)
```

Multiple Linear Regression

Goal: Build a Predictive Model (Regression) to estimate energy expenditure.

Analytical Approach:

- Model: Multiple Linear Regression (MLR).
- Input Features: 4 selected variables (Session Duration, Experience Level, Water Intake, and Workout Types).
- Validation: 80/20 Train-Test Split with Cross-Validation.

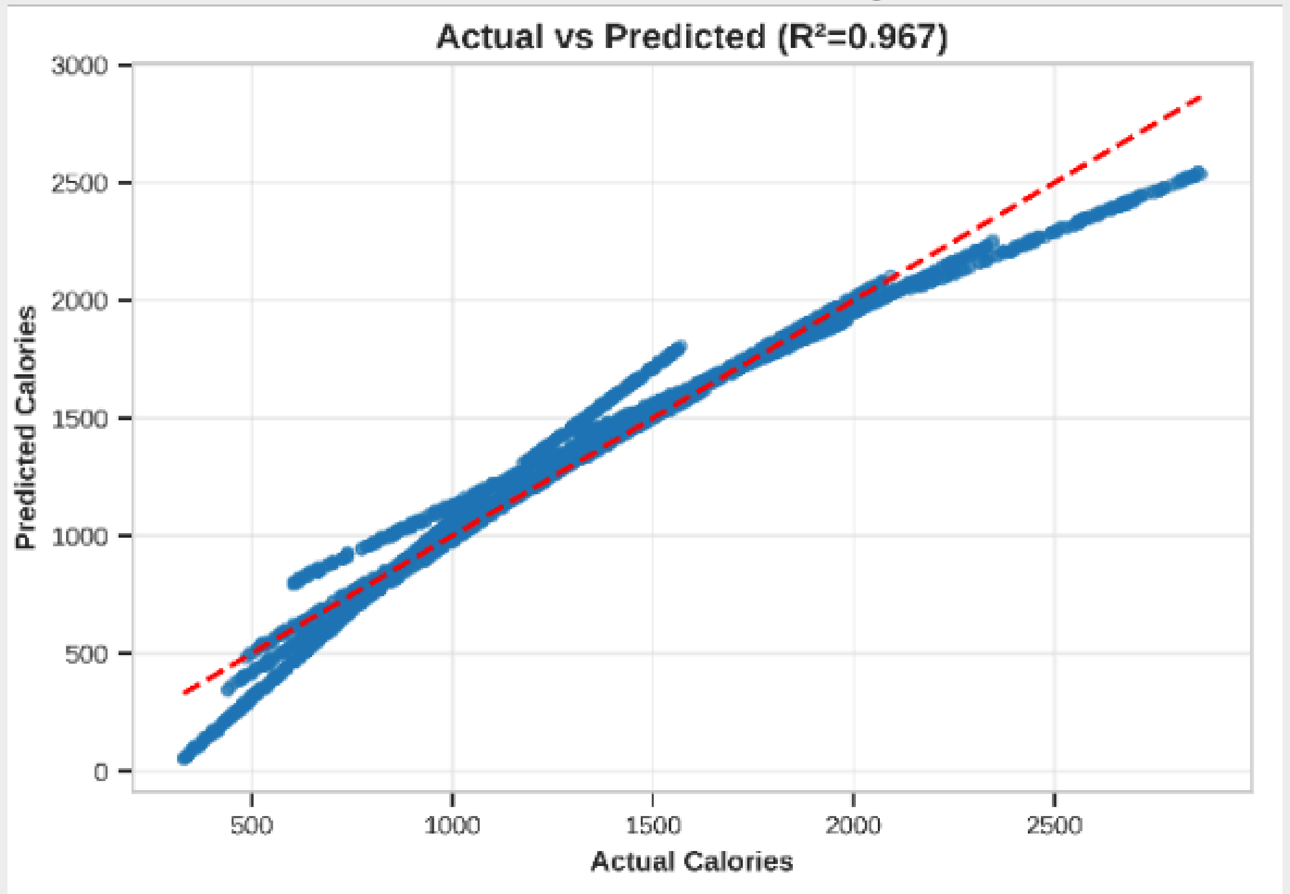


Multiple Linear Regression

Model Reliability (Performance)

Key Insight: The model demonstrates exceptional predictive accuracy, explaining nearly 97% of the variance in calorie expenditure.
Performance Metrics:

- Coefficient of Determination (R2): 0.968 (High Goodness-of-Fit).
- RMSE: 174.40 (Acceptable margin of error given the high range of calorie values).
- Stability: Cross-validation scores confirm robust generalization to new data.



Multiple Linear Regression

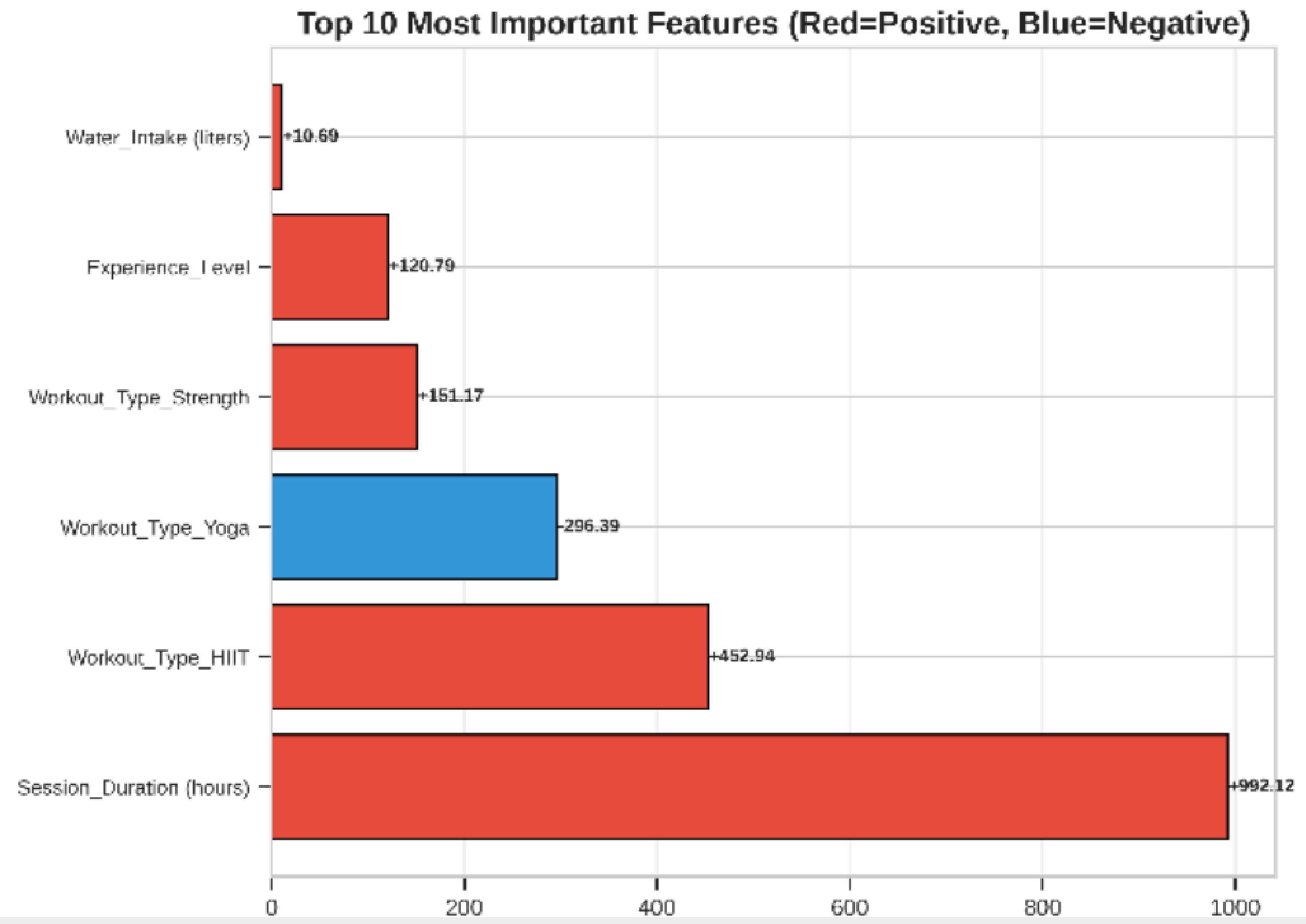
Feature importance and Coefficient analysis

Key Insight:

- Workout Duration and Intensity are the primary drivers of energy expenditure, significantly outweighing other factors.

Critical Findings:

1. Dominant Factor: Session_Duration (+992.12).
   - Interpretation: Holding all else constant, each additional hour of training increases expenditure by approximately 992 calories.
2. Intensity Impact (Workout_Type): HIIT provides a substantial metabolic boost compared to the baseline.



Multiple Linear Regression

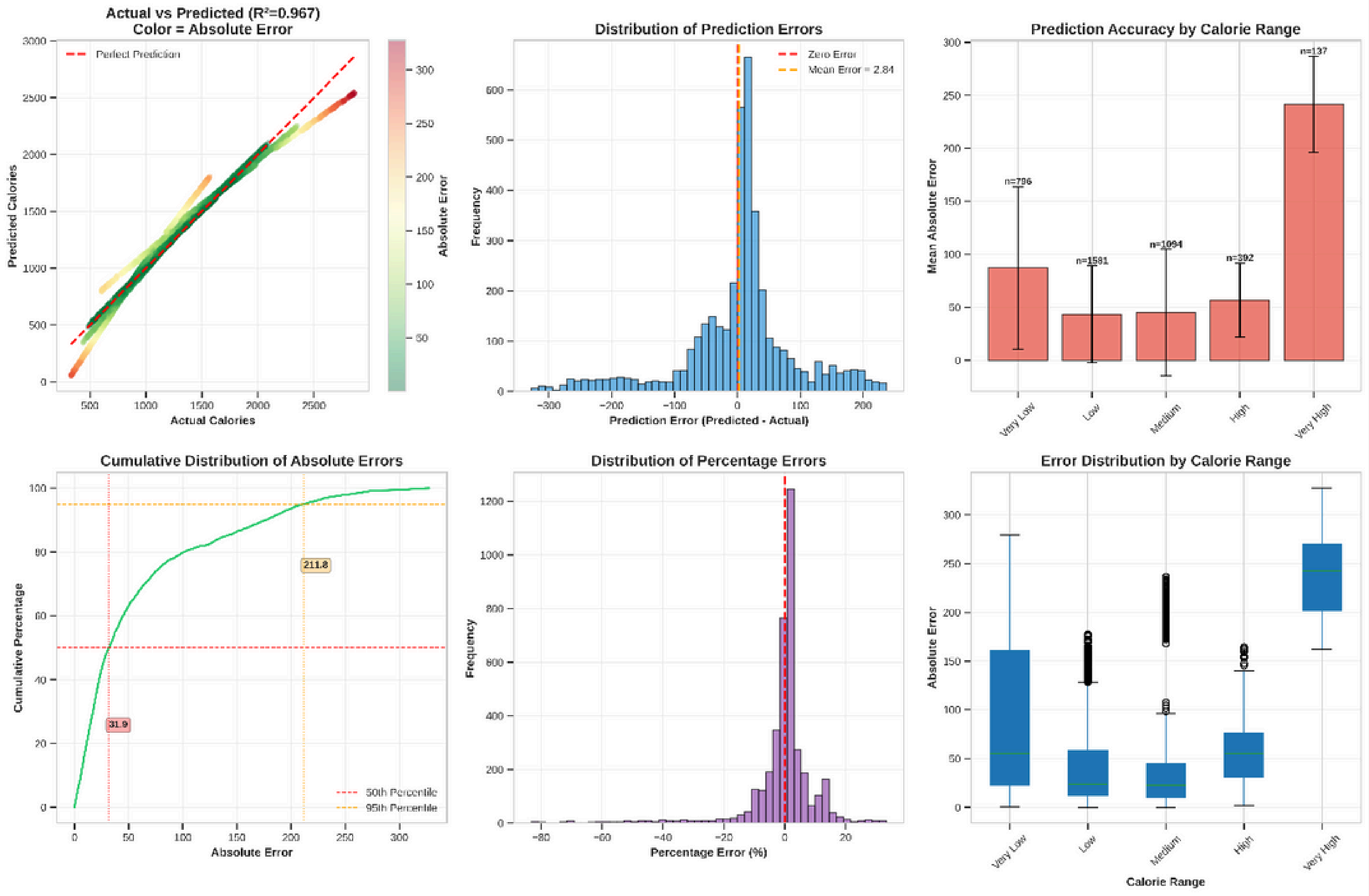
Predictive Capability in practice

Key Insight:

- The model effectively distinguishes between high-performance and low-intensity sessions in real-world scenarios.

Demonstration:

- A side-by-side comparison of "True Values" vs. "Predicted Values" for 5 random test samples shows minimal deviation.
- The model correctly identifies high-burn sessions (e.g., HIIT > 1hr) vs. low-burn sessions (e.g., Yoga < 1hr).



Multiple Linear Regression

Conclusions

Optimization Strategy:

- To maximize caloric output, training protocols should prioritize Duration followed by High-Intensity modalities (HIIT).

Answer for Research Question 2

Which factors are the strongest predictors of "Calories Burned", and can we accurately estimate energy expenditure?

Answer: Top 3 factors ranked by importance

1) Session duration
2) Workout type
3) Experience level

Supporting Evidence

- Correlation Analysis
- Regression Model

Conclusion

Research question 1:
Does Gender significantly affect Workout Type selection?

Result: The hypothesis that "Gender dictates exercise choice" is **REFUTED**.

Research question 2:
Which factors are the strongest predictors of "Calories Burned", and can we accurately estimate energy expenditure?

Top 3 factors ranked by importance:
1) Session duration
2) Workout type
3) Experience level

Thank you